Science Olympiad Thermodynamics test KEY
Kenston Invitational 1/13/18

1. B
2. A
3. D
4. C
5. C
6. B
7. A
8. A
9. B
10. D
11. A
12. B
13.
    a. 2nd law (heat flows hot to cold)
    b. 1st (COE)
    c. 0th (thermal equil)
    d. 2nd law (heat flows hot to cold)
    e. 2nd law (friction, heat transfer, increased disorder/entropy)
14. 25 degrees C
15. -40
16. 617.67 degrees R
17. Latent heat
18. Specific heat
19. $|W| = |P\Delta V| = (2.00\ atm) \times (1.5\ L) \times \frac{10^5 Pa}{1\ atm} \times \frac{0.001\ m^3}{1\ L} = 3,000\ J$